

Adarsh Dwivedi

AI Engineer

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Final-year CS engineer shipping production AI with evaluation wired in from the start. At Deloitte, took a LangGraph agent from in-request execution to a self-hosted agent server with durable PostgreSQL state, proven live on SAP at 8 of 8 tasks. Across 32 public repositories, most of them deployed and free to use, every reported number is reproducible from the repository, and three of them ship as installable packages on PyPI.

EXPERIENCE

Deloitte India, Digital Excellence Centre

Jun 2026 - Aug 2026

AI Engineer (Agentic AI) Intern

Gurugram, India

- Migrated SAP Agentic, Deloitte's system that automates SAP configuration, from LangGraph running inside a FastAPI request handler onto a self-hosted Aegra agent server, with threads, runs and checkpoints in PostgreSQL inside the firewall at no licence cost.
- Proved the new path on the live SAP tenant: a Bank Accounting goal was planned into 8 configuration tasks, executed in a real browser through Playwright MCP, recovered from a duplicate-entry conflict and a mis-filled field, and finished 8 of 8 in about 15 minutes.
- Built capability in five layers on a multi-agent travel planner, from a single agent to a supervisor with three specialists, a human approval gate, real browser tools over MCP and serving through Aegra, proving each layer before starting the next.

PROJECTS

CacheLLM, Semantic Caching Proxy for LLM APIs (Published on PyPI) | [Live](#) | [GitHub](#)

Python, FastAPI, Embeddings, Redis

- Built a drop-in semantic cache keeping the OpenAI request and response shape including streaming, routing each model to the host that serves it across OpenAI, Gemini, Groq, Claude, Grok, Ollama and Bedrock.
- Matched exact-hash first then cosine similarity over normalised embeddings, with the safe threshold calibrated per embedding model rather than assumed, reaching 77% hit rate and 78% cost reduction.
- Held zero wrong answers on new questions by namespacing on model, system prompt and sampling parameters, and refusing to cache any reply truncated by max_tokens.

Vayu, National Air-Quality Intelligence and Enforcement Platform (Deployed on Google Cloud Run) | [Live](#) | [GitHub](#)

LightGBM, CNN-LSTM, DuckDB, Python

- Enforcement targets the dirtiest sensor, not the site where action pays most. Forecast air quality with LightGBM on station data plus a CNN-LSTM over a 15,360-cell satellite grid, then ranked interventions by Gaussian-plume modelled return.
- Kept PyTorch out of the API request path entirely, training the CNN-LSTM offline and serving from DuckDB as one embedded columnar file, so scoring is fast without a model server in the loop.
- Verified that dispatched interventions actually worked using difference-in-differences against control regions, rather than reporting any post-intervention drop as proof of effect.

VaidyaMitra, Privacy-First Clinical GenAI Assistant | [Live](#) | [GitHub](#)

AWS Bedrock Nova, FastAPI, Next.js

- Masked every identifier before it reaches the model across a 38-endpoint API, so a hosted model never sees patient-identifying text.
- Benchmarked four Bedrock models on latency and multilingual quality to select Nova Pro at 605 ms against alternatives above 1,050 ms, and set maxTokens explicitly to avoid quota-reservation throttling.
- Matched prescriptions to Jan Aushadhi generics with substitution-safety warnings, surfacing savings up to 93.7% across 15 Indian languages.

TECHNICAL SKILLS

Generative AI: LangGraph, LangChain, RAG, Prompt Engineering, Vector Search, Embeddings, Evaluation and Calibration, Amazon Bedrock, Gemini, OpenAI, Groq, Ollama

Agentic AI: LangGraph, MCP (Model Context Protocol), Multi-Agent Orchestration, Tool Calling, Human-in-the-Loop, Agent Protocol, Aegra, Playwright MCP, LLMOps

Machine Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, Pandas, NumPy, Feature Engineering, Cross-Validation, Calibration, SHAP, A/B Testing, Statistics

Backend and Infrastructure: FastAPI, Pydantic, Node.js, Express, REST APIs, PostgreSQL, MongoDB, DuckDB, Next.js, React, Streamlit

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL

ACHIEVEMENTS

- Published three packages to PyPI, cachellm-proxy, creditsetu and medguardx-core, each installable, documented and covered by tests.
- KSP Datathon 2026 finalist: reached Level 2 with Kadi, a crime analytics platform built end to end over 50 days and deployed on Zoho Catalyst for the Karnataka State Police.

EDUCATION

The LNM Institute of Information Technology, Jaipur

2023 - 2027

B.Tech, Computer Science and Engineering

CGPA: 7.11 / 10

LEADERSHIP

- Coordinator, Debate Society, LNMIIT Jaipur (May 2025 - Apr 2026): led a 150+ member society across debates, discussions and inter-college competitions.
- Secretariat and Organizer, LNMIIT Model United Nations (Aug 2025 - Jan 2026): organised a large-scale MUN conference, coordinating delegates, committees, schedules and logistics.